

Friction Imbalance Model - Summary

Model Idea:

Pebbles on curling ice are created by spraying water droplets on the ice beforehand. When the curling stone goes over each pebble, it heats the pebble up due to friction and pressure. The pebbles then cool down when no longer in contact with the running band of the curling stone.

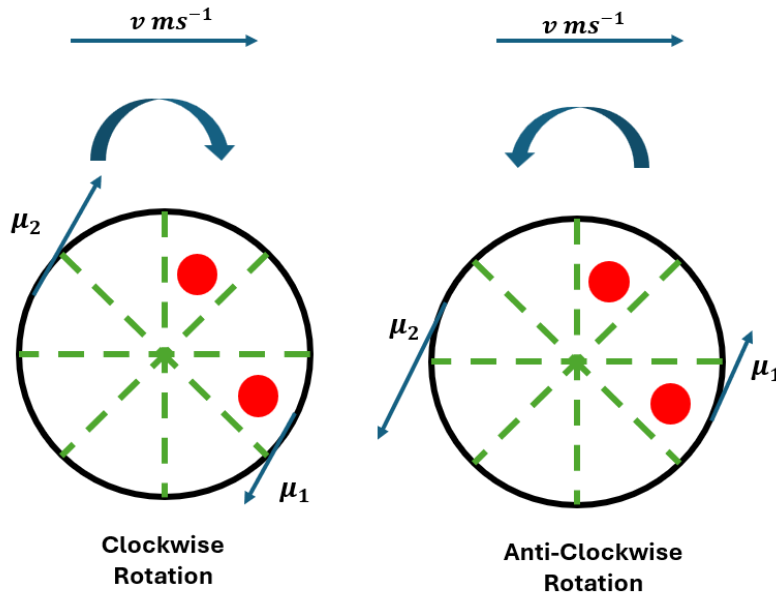


Figure 1: A diagram showing the curling stone split into 8 angular sections, with 2 pebbles under the running band. μ_1 and μ_2 are also drawn on the diagram next to the direction of their corresponding forces. In this case, $\mu_2 > \mu_1$. Both clockwise and anti-clockwise rotation is shown.

In this model, I am splitting the curling stone into n angular sections as shown in Figure 1, where n is an even integer. The coefficient of friction is then determined using the equation:

$$\mu = \frac{2}{pv} \left[\frac{k_g \Delta T_g}{\sqrt{\pi \kappa_g t_p}} + \frac{k_i \Delta T_i}{\sqrt{\pi \kappa_i t_b}} \right] \quad \text{eq. [1]}$$

Where μ is the coefficient of friction, v is the translational velocity of the curling stone, k_g is the thermal conductivity of the granite with a value of $2.94 \text{ W m}^{-1} \text{ K}^{-1}$, ΔT_g is the difference between the running band temperature and the freezing point temperature. κ_g is the thermal diffusivity of the granite with a value of $1.49 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$, k_i is the thermal conductivity of the ice with a value of $2.3 \text{ W m}^{-1} \text{ K}^{-1}$. κ_i is the thermal diffusivity of ice with a value of $1.23 \times 10^{-6} \text{ m}^2 \text{ s}^{-1}$, t_b is the contact time between the running band and a point on the pebble.

p is the ice hardness with an equation of $p = 14.7 - 0.6T$ with units of MPa. t_p is equal to $t_p = 2r_p/v$ where r_p is the radius of the pebble. ΔT_i is the difference between the temperature of the ice temperature and the ice melting point with an equation of $\Delta T_i = 0 - T_p$ where T_p is the temperature of the pebble, all temperatures in $^{\circ}\text{C}$. I am using a constant value of $t_b = \frac{w}{v}$ where w is the running band width, t_b can have a range of values.

Then, the coefficient of friction is then calculated for the opposite point on the curling stone as shown in Figure 1. The difference between the two coefficients of friction values is then calculated using the equation:

$$\Delta\mu = \mu_1 - \mu_2 \quad \text{eq. [2]}$$

Where $\Delta\mu$ is the difference in the coefficient of friction values, μ_1 is the coefficient of friction is the coefficient at one point on the running band and μ_2 is the coefficient of friction on the opposite side of the running band.

Then, this change in mu is converted into a force using the equation:

$$\Delta F = \Delta\mu N \quad \text{eq. [3]}$$

Where ΔF is the change in friction between the two sides of the stone and N is the normal force. This is then converted into X and Y components, then this process is repeated for all the other angular sections so that there are $\frac{n}{2}$ forces with X and Y components. These forces are summed to get the overall force on the stone.

Heating of Pebbles:

When the pebble is in contact with the running, it heats up using the equation:

$$dT = \frac{\mu_0 g v dt}{N_p c} \quad \text{eq. [4]}$$

Where μ_0 is the assumed baseline coefficient of friction with a value of 0.01, g is 9.81ms^{-2} , dt is the timestep, N_p is the number of pebbles touching the entire running band and c is the specific heat capacity of ice with a value of $2,100 \text{J/kg}^\circ\text{C}$ and dT is the change in temperature in one timestep.

Cooling of Pebbles:

When the pebble is no longer in contact with the running band, it cools down using the equation:

$$dT = \frac{-2k_i \Delta T_i}{\sqrt{\pi \kappa_i t_b}} \times \frac{A dt}{mc} \quad \text{eq. [5]}$$

Where A is the contact area of one pebble and m is the mass of one pebble.

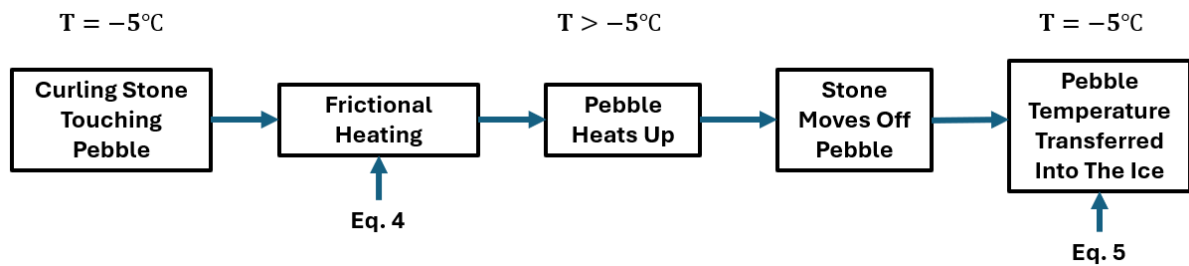


Figure 2: A diagram showing how the process of pebble heating and cooling happens, with temperatures labelled as well as the corresponding equations.

The equation for the coefficient of friction is from (Lozowski et al., 2015) and the equations for pebble heating and cooling are also adapted from this paper.

Assumptions:

All pebbles have the same contact area and radius.

All pebbles touching the running band are heated at the same rate and cool down at the same rate once no longer touching the running band.

Assuming all excess heat from pebbles when cooling is conducted into the surrounding ice sheet.

Using a constant value of t_b .

Predictions:

This model will produce a final y displacement that is less than 10cm.

The direction of the curl will be random as the pebbles are randomly placed on the icesheet.

Results:

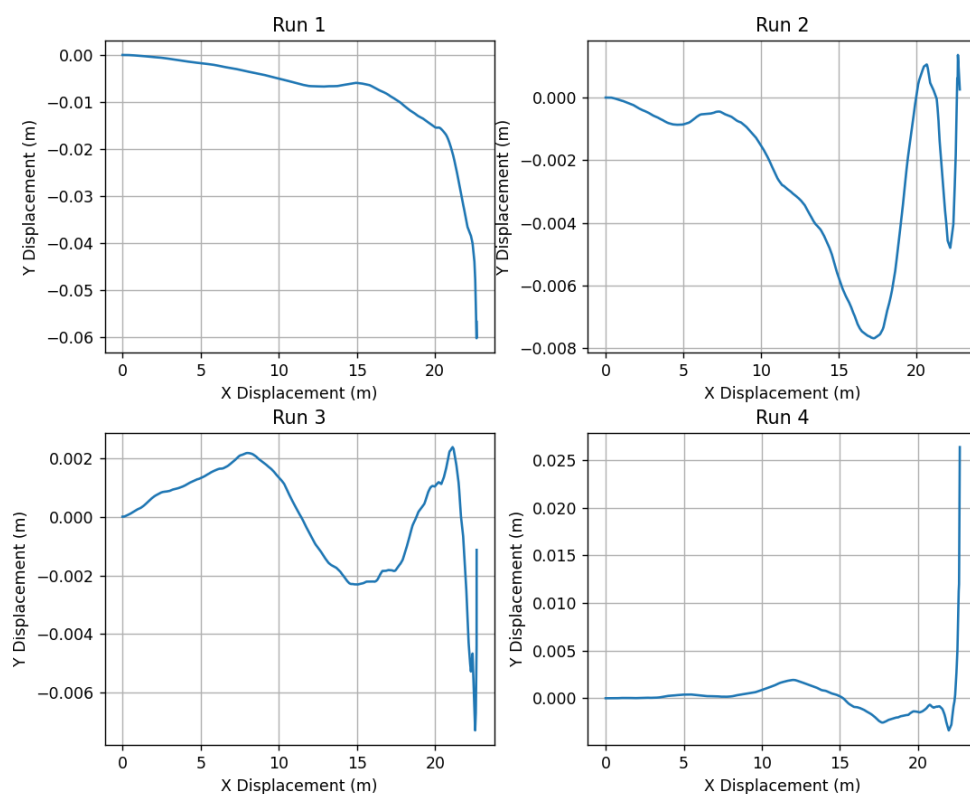


Figure 3: 4 graphs showing the x and y displacements of the curling stone moving across 4 different random pebble fields, plotted in python.

20 y displacement readings for a clockwise rotation and 20 y displacement readings for an anti-clockwise rotation were taken. The results of this are as follows:

- The average curl of the clockwise rotation is 8.14mm and the average curl of the anti-clockwise rotation is -1.96mm
- The average magnitude of curl for the clockwise rotation is 18.2mm and the average magnitude of curl for the anti-clockwise rotation is 29.2mm
- Clockwise rotation had 8 negative y displacements and 12 positive y displacements

- Anti-clockwise rotation has 11 negative y displacements and 9 positive y displacements
- The motion of the curling stone does not follow one perfect curve. For example, can have a negative y displacement at 12m and a positive one at 25m

Conclusion:

Overall, the results of this model show that there is no real bias in the direction the stone curls, in this particular model, due to thermal changes on the icesheet. The average magnitude of curl is 23.7mm with a standard deviation of 21.3mm. Due to the large standard deviation, it is not possible to predict the magnitude or direction the curling stone will curl in, due to thermal changes on the icesheet, for any specific run of the model. From the plots above, the path of the curling stone in this model is not one perfect curve.

Future Improvements:

- Variable t_b value instead of using a constant value
- Accounting for scratches and pivoting